

PROBLEM STATEMENT

Digital Platform to Crowdfund Societal Challenges

Industry Context

Society faces many challenges such as poor waste management, damaged roads, water leakage, environmental pollution, public safety concerns, inadequate public facilities, and other local community issues. These problems may occur frequently in residential areas, educational institutions, public places, and urban environments. Identifying and resolving such issues requires active participation from citizens, local authorities, and other responsible organizations.

In many existing systems, citizens report societal problems through manual complaints, phone calls, emails, social media, or separate government portals. These methods may make it difficult to collect, organize, prioritize, and track community issues in a centralized manner. Citizens may also have limited information about whether their complaints have been received, assigned, investigated, or resolved.

The absence of an effective centralized crowdsourcing platform can lead to:

- Delay in reporting societal problems
- Difficulty in identifying problems at an early stage
- Lack of proper communication between citizens and authorities
- Duplicate complaints for the same issue
- Difficulty in prioritizing critical community problems
- Limited transparency in complaint handling
- Lack of real-time status tracking
- Difficulty in identifying areas with frequently occurring problems
- Inefficient allocation of resources
- Reduced citizen participation in community improvement

With the increasing use of smartphones, internet services, and digital platforms, citizens can actively contribute information about problems in their surroundings. A crowdsourcing-based platform can collect these reports from multiple users and organize them in a centralized system.

The collected information can include the type of societal challenge, location, description, images, date, severity, and user-provided details. The system can analyze and categorize these reports to help responsible authorities identify important issues and take appropriate action.

Therefore, there is a need for a digital platform that enables citizens to report societal challenges, allows multiple users to contribute information, categorizes and prioritizes reported issues, tracks their resolution status, and provides useful information to authorities and the community.

Problem Statement

Design and develop a **Digital Platform to Crowdsource Societal Challenges** that enables citizens to report, share, track, and monitor societal problems occurring in their communities.

The system should allow users to submit information about societal challenges such as waste accumulation, road damage, water leakage, pollution, public infrastructure problems, and other community-related issues. Users should be able to provide relevant details such as problem description, category, location, severity, date, and supporting images wherever applicable.

The collected reports should be stored and organized in a centralized database. The system should provide suitable mechanisms for categorizing reported challenges and identifying duplicate or similar reports. This will help avoid repeated complaints and improve the organization of community information.

The platform should provide a dashboard that displays important information such as newly reported challenges, pending issues, resolved issues, high-priority problems, frequently reported categories, and affected locations. This will help administrators and responsible authorities understand the current situation and manage reported problems efficiently.

When a critical or high-priority societal challenge is reported, the system should provide notifications or alerts to the responsible stakeholders. The system should also allow administrators to assign reported issues to appropriate departments or responsible persons for further action.

The platform should maintain the complete history of reported challenges, including their reporting date, status, assigned authority, actions taken, and resolution details. Citizens

should be able to track the progress of their submitted reports and receive updates when the status changes.

The system should also provide visualizations and reports to identify frequently occurring societal challenges, affected areas, problem categories, and resolution trends. This information can support authorities in understanding recurring community problems and planning suitable preventive measures.

The proposed solution should be designed with security, scalability, reliability, usability, maintainability, and efficient data management in mind. It should support a large number of users and reports while providing timely information to citizens and responsible authorities.

Objectives

The system should:

- To provide a digital platform for citizens to report societal challenges.
- To collect information about community problems from multiple users.
- To store reported challenges in a centralized database.
- To capture important details such as category, location, description, severity, date, and supporting images.
- To categorize reported societal challenges based on their type.
- To identify duplicate or similar complaints.
- To prioritize reported problems based on their severity and importance.
- To monitor the status of reported societal challenges.
- To assign reported issues to appropriate authorities or responsible stakeholders.
- To provide notifications for important or high-priority challenges.
- To allow citizens to track the progress of their submitted reports.
- To maintain historical records of reported and resolved challenges.
- To generate reports and visualizations for community problems.
- To identify frequently occurring challenges and affected locations.
- To improve communication and collaboration between citizens and authorities.

- To support faster and more effective resolution of societal challenges.

Expected Outcomes

1. Develop an end-to-end **Digital Crowdsourcing Platform** for collecting, managing, analyzing, and tracking societal challenges.
2. Provide a user-friendly GUI/dashboard for citizens, administrators, and responsible authorities to report, monitor, and manage community problems.
3. Maintain information such as problem category, location, description, severity, reporting date, images, status, assigned authority, resolution details, and historical records.
4. Enable citizens to submit societal challenges digitally and contribute information about problems occurring in their surroundings.
5. Provide categorization and prioritization of reported challenges to help authorities focus on important and urgent issues.
6. Provide mechanisms for identifying duplicate or similar complaints and improving the organization of reported information.
7. Provide real-time or timely status updates so that citizens can track the progress of their submitted challenges.
8. Generate dashboards, reports, and visualizations showing problem categories, frequently affected areas, pending issues, resolved issues, and resolution trends.
9. Provide notifications and alerts for high-priority or critical societal challenges.
10. Allow users and administrators to view, analyze, and export relevant reports and collected data in suitable formats such as CSV and JSON.
11. Apply important software engineering principles such as security, scalability, maintainability, reliability, usability, and efficient data management.
12. Demonstrate proper requirements analysis, system design, SDLC selection, database design, user-interface development, testing, risk analysis, limitations, and future enhancements.